

THE DAYNIGHT BELT

A3 Booklet · A Time Contract with the City: a 24h Youth-Friendly Public Space Belt



Dawn = cool daylight blue (co-create / sport / open-source)

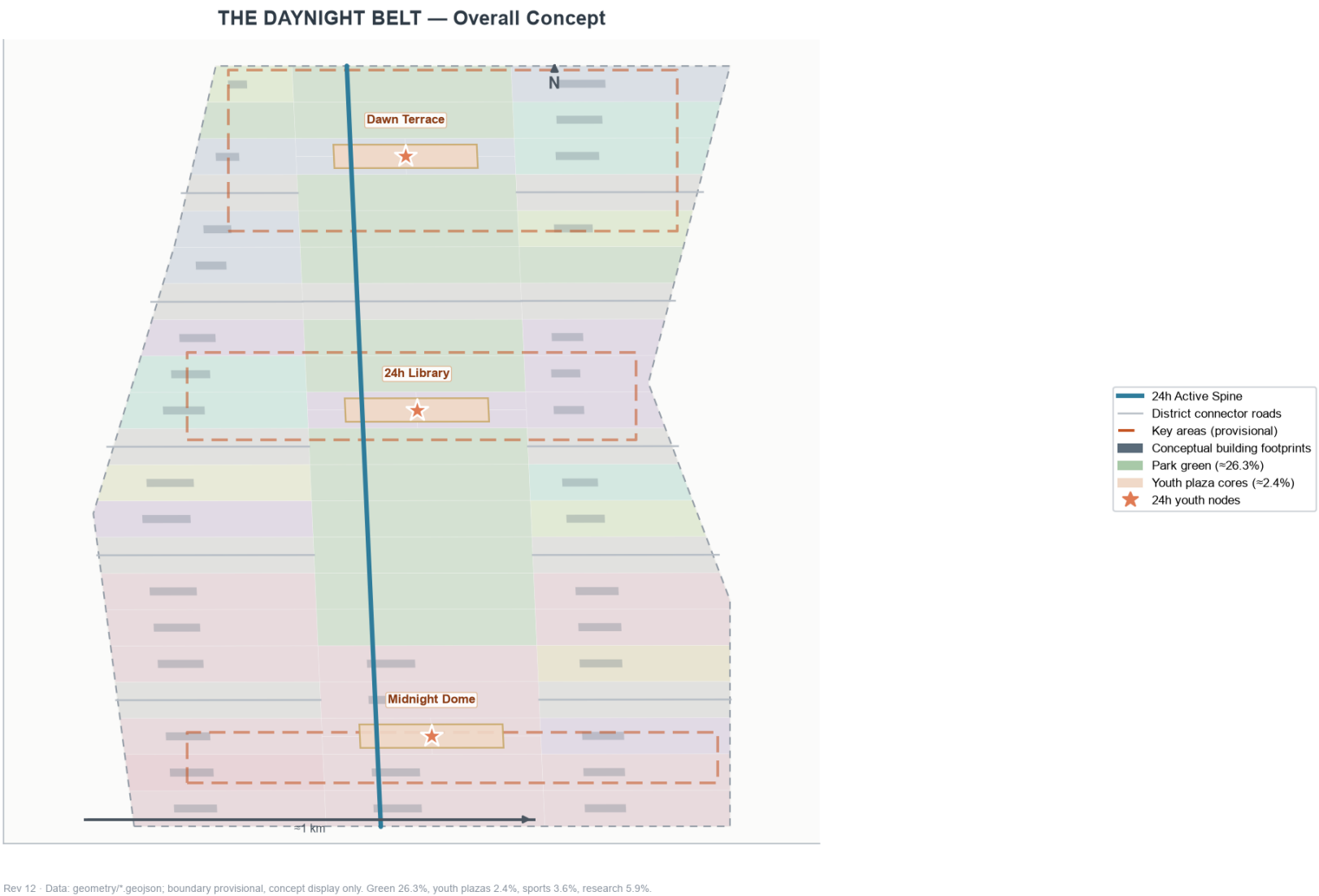
Night = warm amber (consumption / nightlife / social)

Junction = youth all-day node

Overall concept: the terminator line (day/night divide) as a temporal motif organizing the Jing-Zhang Heritage Park belt into a 24h youth-friendly public space belt.

Submitted boundary: provisional ≈11.4 km2; key areas ≈368.4 ha (three).

This package is conceptual; all spatial proposals are reference schemes pending official boundaries and regulatory conditions.



01 Overall Concept and Spatial Structure

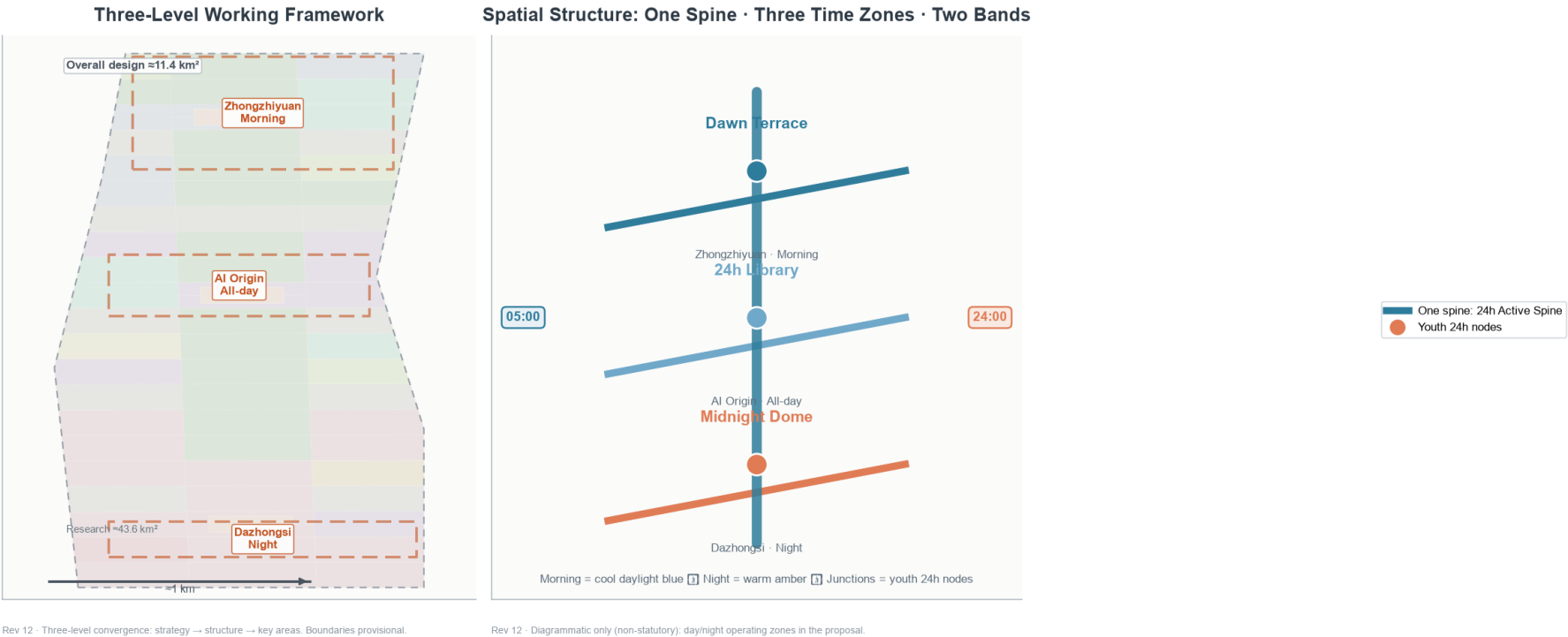
Overview and Spatial Structure

One Spine · Three Time Zones · Two Color Bands

- One Spine: Jing-Zhang Heritage Park 24h Active Spine (green belt + slow mobility + lighting grading) linking three youth all-day nodes.
- Dawn Terrace · Zhongzhiyuan (N, morning): youth co-create workshops, sports, open-source stations — morning to noon.
- 24h Library · AI Origin (C, all-day): youth culture, study, sharing, developer salons — all day.
- Midnight Dome · Dazhongsi (S, night): youth consumption, night-life market, social — noon to night.
- Two color bands: cool daylight blue (#2E7C9B) and warm amber (#E07B54) graded along the terminator; junctions become node circles.
- Time contract: railways once compressed 'time over the mountains'; AI cities redistribute 'time of the day'; youth are the measure of a 24h city.
- Public-life first: green 26.3%, youth plaza cores 2.4%, sports 3.6%, community service 1.8% (distinct from industry-led schemes).

Naming System and Logo Direction (agent.1)

Primary name: The Daynight Belt. Logo direction: one terminator-gradient two-band stripe,
cool daylight blue meeting warm amber, the junction dissolving into a circle, extended into the Chenhun Festival identity.



02 Three Key Areas — Detailed Design

Key Area Detailed Design

Zhongzhiyuan · Morning Zone (≈192.1 ha announced)

- Position: youth co-create & sport belt; Dawn Terrace core; research/sports/green complex
- Renewal: retrofit existing buildings + pilot facilities (concept)
- Scenarios: dawn run companion (S01), motion health recognition (S02), open-source morning desks
- Risk: complex ownership, parcel-level confirmation; regulatory data pending

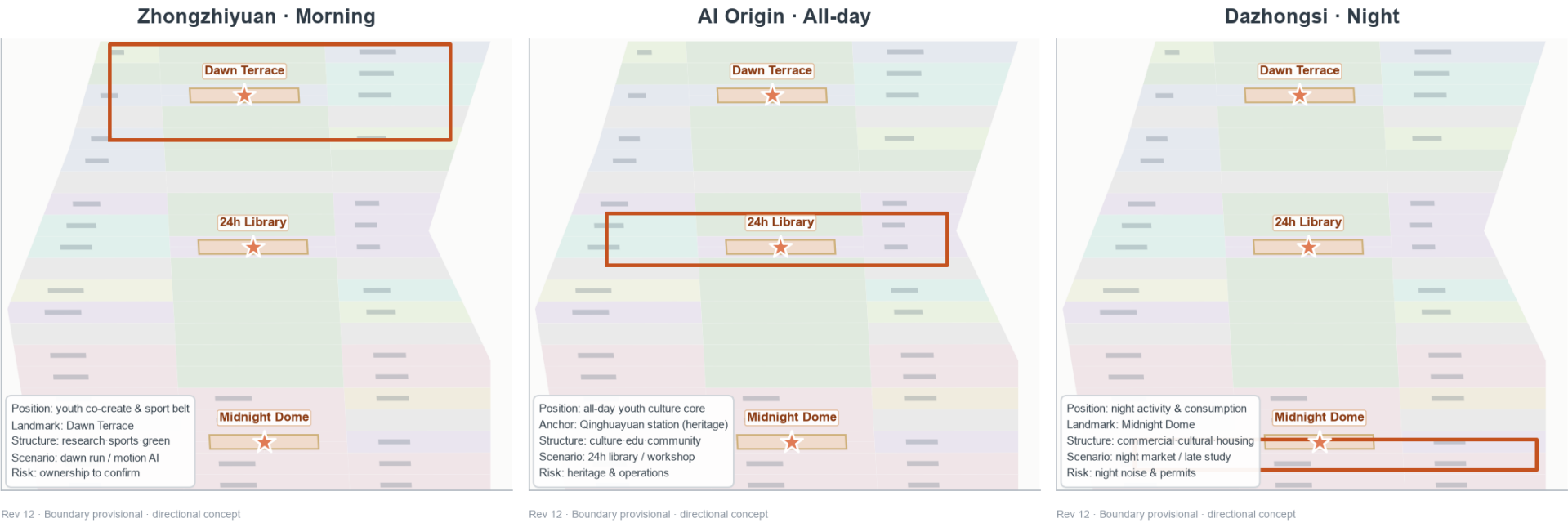
AI Origin Community · All-day (≈104.3 ha announced)

- Position: all-day youth culture core; Qinghuayuan Station site as cultural anchor (per heritage authorities)
- Structure: culture (≈74 ha concept) + education + community service around the 24h Library plaza
- Scenarios: 24h library (S04), developer late-night salon (S12), shared-facility booking agent (S03)
- Risk: heritage and university ownership highly sensitive

Dazhongsi · Night Zone (≈72.0 ha announced)

- Position: night activity & consumption belt; Midnight Dome core
- Renewal: youth-friendly commercial street renewal (concept, no demolition assumptions)
- Scenarios: night-life market (S07), adaptive night lighting (S05), late-night study pods (S09)
- Risk: night noise & operating permits need market and planning double confirmation

Three Key Areas · Youth All-day Nodes



03 Youth Personas, Scenarios and Landmarks

Youth Personas, Scenarios and Landmarks

Six youth personas: Students P1 / Startups P2 / Developers P3 / Night owls P4 / Young parents P5 / Silver-age visitors P6

12 youth scenario cards (incl. 3 test/validation scenarios)

S01 Dawn-run companion (test/validation) | S02 Morning motion health recognition (test/validation) | S03 Shared-facility booking agent (test/validation)
S04 24h library | S05 Adaptive night lighting | S06 Late-night study pod | S07 Night-life market | S08 Youth co-create workshop
S09 Parent-child dawn station | S10 Skate plaza | S11 Silver-age dawn stroll | S12 Developer late-night salon
All scenarios are conceptual — not deployed, not authorized, not operational; each has a manual equivalent path and human shut-down condition.

Three youth pilgrimage landmarks (agent.4, conceptual)

- 1 Dawn Terrace (Zhongzhiyuan Youth Plaza) · dawn exercise, morning market and 'first light' viewing deck
- 2 24h Library (Origin Youth Plaza) · all-day study/reading/sharing + developer star registry
- 3 Midnight Dome (Dazhongsi Youth Plaza) · night-life main stage, the echo between technology and life

Core Metrics Recalculation & Evidence Chain

Metric	Value	Formula	Evidence
Green ratio	26.3%	park green + submitted boundary area	land_use.geojson / metrics.json
Youth plaza core ratio	2.4%	plaza core + submitted boundary area	land_use.geojson / metrics.json
Youth sports land	3.6%	0805 area + submitted boundary area	land_use.geojson / metrics.json
Community service land	1.8%	0702 area + submitted boundary area	land_use.geojson / metrics.json
Research land	5.9%	0802 area + submitted boundary area	land_use.geojson / metrics.json
Commercial land	23.6%	05 area + submitted boundary area	land_use.geojson / metrics.json
Road land ratio	16.4%	1207 area + submitted boundary area	land_use.geojson / metrics.json
Building footprint	2.6% (34)	conceptual footprints + boundary area	buildings.geojson / metrics.json
24h active spine	≈9.7 km	projected length (EPSG:4548)	roads.geojson / metrics.json
Key-area area	192.9 / 104.3 / 72.0 ha	Zhongzhiyuan / Origin / Dazhongsi	key_areas.geojson
Youth 24h nodes	3	Dawn Terrace / 24h Library / Midnight Dome	public_space.geojson

Unknown: FAR / building height / per-capita public space / night lighting coverage — missing official baseline, registered as unknown (metrics.json)

Recompute: Shapely + EPSG:4548

Rev 12 - Working values on provisional boundary, recompute when official polygons land.

04 Core Metrics Recalculation

Core Metrics Recalculation

Metric	Value	Formula
site_area_sqm	1.14128e+07 sqm	<code>polygon_area(submitted_site_boundary)</code>
green_ratio	0.263 ratio	<code>green_area / site_area</code>
public_space_ratio	0.024 ratio	<code>plaza_area / site_area</code>
sport_land_ratio	0.036 ratio	<code>sport_land_area / site_area</code>
road_ratio	0.164 ratio	<code>road_area / site_area</code>
building_footprint_ratio	0.026 ratio	<code>building_footprint_area / site_area</code>
spine_length_m	9723.47 m	<code>projected_length(design_role=north_south_spine, EPSG:4548)</code>
key_area_count	3 count	<code>count(key_area_features)</code>
zhongzhiyuan_area_sqm	1.9292e+06 sqm	<code>polygon_area(zhongzhiyuan_ai_acceleration_area)</code>
beijing_ai_origin_area_sqm	1.04324e+06 sqm	<code>polygon_area(beijing_ai_origin_community)</code>
dazhongsi_area_sqm	720454 sqm	<code>polygon_area(dazhongsi_ai_industry_cluster)</code>

Unknown metrics (pending official data): floor_area_ratio / building_height_control_m / per-capita public space / night lighting coverage



Rev 12 - Centerlines conceptual, not road redlines. Spine ≈9.7 km, conceptual network ≈23.6 km.

05 Renewal Projects, Phasing and Compliance

Renewal Projects, Phasing and Compliance

Three phases with evaluation gates (youth all-day nodes first)

- P1 near-term: Origin all-day core (cultural repair near Qinghuayuan Station + 24h Library plaza), spine mid-section 24h slow-mobility connection.
- P2 mid-term: Zhongzhiyuan morning zone (retrofit stock buildings into youth co-create carriers), Dawn Terrace and youth sports fields.
- P3 long-term: Dazhongsi night zone (youth-friendly commercial street renewal), Midnight Dome and night lighting grading.

Long-term operations (agent.6, conceptual)

- Annual Chenhun (Daynight) Festival: dawn-run month / 24h reading week / night-vitality season (Sep–Oct peak), an all-time activity system
- Youth co-creation fund (concept): funding shared facilities, night-life formats and community co-creation proposals (review–pilot–evaluate–expand or withdraw)
- All-time safety: night lighting grading, aggregate-only surveillance, human staffing; any scenario can be shut down as a whole

Compliance notes

- Only official public announcements, the cleared taskbook, public policies and provisional geometry are used; no internal data.
- COMMUNITY-DISPLAY-ONLY; the logo is a textual direction, not a trademark registration.
- All spatial proposals are conceptual and subject to final human and professional judgment; night-life formats require operating permits.